

Research Article

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Extracting Inductive Bias Using Generative Models

<https://doi.org/10.1515/sample-YYYY-XXXX>

Received Month DD, YYYY; revised Month DD, YYYY; accepted Month DD, YYYY

Abstract: Inductive biases shape how neural networks generalize from limited data, yet encoding them in network connectivity often relies on domain expertise. This paper presents a generative approach that draws on related tasks to discover shared structural constraints. For each task, a generative model approximates a family of hypothesis spaces that admit near-optimal solutions. Intersecting these task-specific families narrows the search to structures compatible with all tasks, automating the discovery of a shared inductive bias (want lossless). Experiments on synthetic tasks demonstrate improved data efficiency and generalization to unseen tasks, supporting the hypothesis that the recovered constraints capture transferable structure.

Keywords: inductive bias, generative model

1 Introduction

TODO

2 Related Works

TODO

3 Preliminary

TODO

4 Method

TODO

5 Experiments

TODO



6 Conclusion

TODO

References

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- [1] Raymond A Yeh, Yuan-Ting Hu, Mark Hasegawa-Johnson, and Alexander Schwing. Equivariance discovery by learned parameter-sharing. In *International Conference on Artificial Intelligence and Statistics*, pages 1527–1545. PMLR, 2022.